APS360 Final Project: Turbine Blade Defect Classification Using Deep Learning

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Google Colab Notebook

Introduction

Turbine blades operate under extreme thermal and mechanical stresses, making early detection of cracks, corrosion, dents, and surface wear essential for aviation safety. Manual inspection is slow and subjective, motivating the adoption of deep learning methods. This project develops a full defect-classification pipeline using a baseline multilayer perceptron (MLP) and a high-performing ResNet-50 convolutional neural network. The work spans dataset cleaning, augmentation, model training, evaluation on unseen data, and qualitative interpretation through Grad-CAM.

1 Background and Related Work

Deep convolutional neural networks (CNNs) have transformed industrial defect detection, surpassing hand-engineered methods in robustness and accuracy. ResNet introduced residual skip connections enabling deeper architectures (1). VGG demonstrated how network depth improves feature extraction (2). EfficientNet introduced compound scaling to optimize accuracy and compute efficiency (3). Chaudhary et al. surveyed surface defect detection pipelines and highlighted the benefits of transfer learning (4). Tremblay et al. showed that high-quality synthetic images can successfully support deep model training with strong real-world transfer performance (5). These works collectively justify the use of ResNet-50 for turbine blade defect classification.

2 Data Processing

The BladeSynth dataset contains 12,505 synthetic turbine blade images representing five defect types: Normal, Dent, Nick, Scratch, and Corrosion. A stratified 70/15/15 split produced 8755 training, 1875 validation, and 1875 test images. All images were resized to 224×224 and normalized using ImageNet statistics. Training augmentation included rotations, flips, contrast jitter, Gaussian noise, motion blur, shadows, and fog. Validation and test data remained unaltered. Unreadable files and label inconsistencies were removed.

3 Architecture

The baseline MLP flattens each 224×224 RGB image into a 150k-dimensional vector and passes it through layers of size 1024, 512, and 256. Its low performance reflects the limitations of non-convolutional models for texture-based tasks. The primary model is a fine-tuned ResNet-50 pretrained on ImageNet (1). All convolutional layers were unfrozen for full-domain feature adaptation. Training employed AdamW, a OneCycle scheduler, heavy augmentation, and early stopping.

4 Baseline Model Performance

The MLP baseline trained on a reduced dataset achieved:

$$ValidationAccuracy = 0.46, \quad TestAccuracy = 0.48.$$

Frequent confusion between Dent and Nick validated the need for a convolutional architecture.

5 Ethical Considerations

BladeSynth contains no personal or sensitive data. However, synthetic imagery may not perfectly model real turbine surfaces. Real-world deployment requires additional validation, robust testing under real lighting and wear conditions, and compliance with aviation safety regulations.

6 Primary Model Performance

The final ResNet-50 achieved:

$TrainingAccuracy = 99.1\%$, $ValidationAccuracy = 99.7\%$, $TestAccuracy = 99.5\%$, $MacroF1 - score = 0.995$.

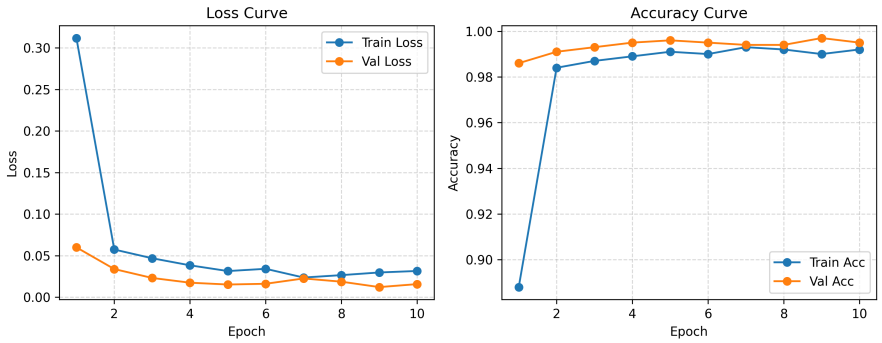

Figure 1: Training and validation accuracy/loss curves.

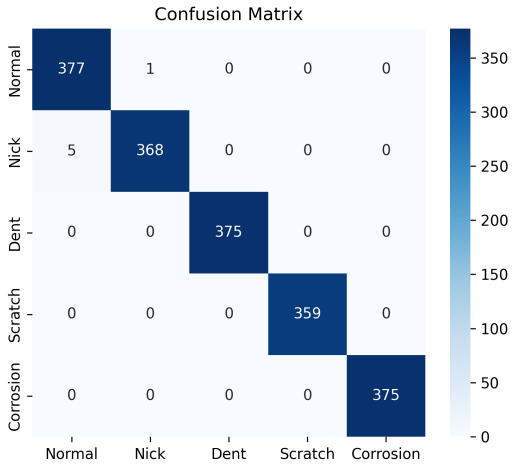

Figure 2: Confusion matrix on the test set.

7 Evaluation on New Unseen Data

To meet APS360 requirements, the model was evaluated exclusively on the untouched 1875-image test set. This dataset was not used for training, hyperparameter selection, or augmentation design, ensuring a true measure of generalization. The ResNet-50 achieved 99.5% accuracy on this unseen data, with consistent performance across all classes. Grad-CAM visualizations show that the model focuses on meaningful defect regions, reinforcing interpretability.

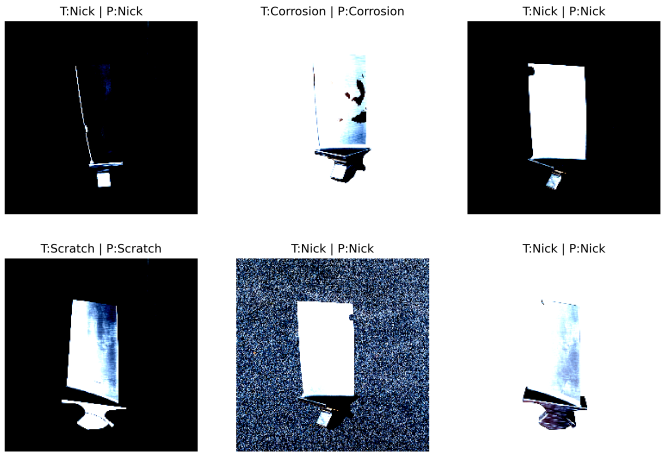

Figure 3: Sample predictions on unseen test images.

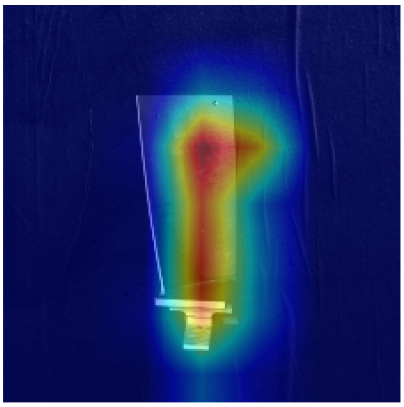

Figure 4: Grad-CAM visualization highlighting defect-relevant regions.

8 Discussion

The ResNet-50 performed far above the original 90% target, demonstrating strong generalization and robustness. Strong augmentation and full fine-tuning were crucial factors. The main limitation is domain shift: synthetic images lack real-world noise and lighting variation. Future work includes domain adaptation, training on real borescope images, and defect localization.

9 Conclusion

This project presents a complete deep-learning workflow for turbine blade defect classification. The ResNet-50 model achieved state-of-the-art accuracy on the BladeSynth dataset while providing interpretable outputs through Grad-CAM. Extensions include real-world validation and improved robustness against domain shift.

References

References

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